

InuaBiz x IntaSend

Payment Integration & Subscription Engine Specification

Comprehensive Blueprint for M-Pesa STK Push, Platform Billing, Webhook Reconciliation & Automated Split Payments

Author:	Zachariah Komu Nyokabi (Full-Stack / AI Engineer)	Integration Target:	IntaSend REST API & Webhooks
Platform Stack:	React, Supabase Edge Functions, PostgreSQL	Document Version:	v1.0 — Production Blueprint

1. Executive Overview & Strategic Choice

IntaSend acts as the central financial gateway for **InuaBiz**. It handles two vital operational flows: collecting monthly platform subscription revenue from vendors (KES 3,000 / month) and processing customer transactions on behalf of micro-vendors.

Why IntaSend over Direct Safaricom Daraja API?

Direct Safaricom Paybill/Till integration requires vendors or platform operators to undergo rigorous compliance checks, business registration certificates, and lengthy approval timelines. IntaSend provides an immediate, plug-and-play M-Pesa STK Push infrastructure, automated webhook retries, built-in card processing, and compliant split payments out-of-the-box.

2. Dual-Flow Integration Architecture

Use Case	Flow Type	Mechanism	Trigger & Outcome
1. Vendor SaaS Billing	Inbound (Platform Revenue)	IntaSend Billing API / M-Pesa STK Push	Triggered at day 14 of trial or billing cycle. Vendor prompts M-Pesa PIN on phone for KES 3,000. Account renewed upon webhook verification.
2. Vendor Sales Processing	Merchant Processing	M-Pesa STK Push / Dynamic QR / Paybill Sync	Customer checks out at POS. Instant STK push sent to customer phone. Funds landed in IntaSend wallet for instant vendor reconciliation.
3. Revenue / Partner Split	Automated Disbursement	IntaSend Split Payments / Transfer API	Automates payouts or fee splits behind the scenes without manual intervention or double-taxation hurdles.

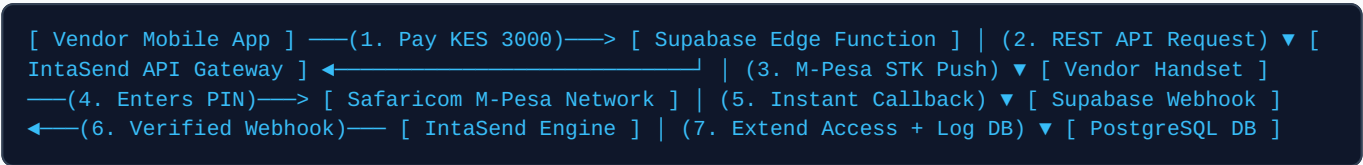
3. Step-by-Step Implementation Workflow

3.1 VENDOR PLATFORM SUBSCRIPTION (KES 3,000 STK PUSH)

When a vendor reaches the end of their 14-day trial or renewal date, the platform triggers an STK Push prompt to their registered phone number.

- Initiate Subscription Prompt:** Front-end calls Supabase Edge Function `/create-subscription-charge` with `tenant_id` and phone number.
- Payload Dispatch:** Edge Function calls IntaSend Checkout API (`https://payment.intasend.com/api/v1/checkout/`).
- M-Pesa Prompt Execution:** Customer receives the system prompt on handset: *"Pay KES 3,000 to InuaBiz Services?"*
- Webhook Reconciliation:** IntaSend sends a signed JSON payload to Supabase Edge Function `/intasend-webhook`.
- Tenant Status Update:** System verifies signature, updates `subscription_status = 'active'`, and extends access for 30 days.

3.2 TECHNICAL SEQUENCE DIAGRAM (TEXT REPRESENTATION)



4. Webhook Security & Payload Specifications

4.1 SECURITY & SIGNATURE VERIFICATION

To prevent spoofed payment requests, all incoming webhooks from IntaSend are cryptographically validated inside our Edge Function before modifying tenant databases.

- Shared Challenge Token:** Stored securely in Supabase Secrets (`INTASEND_WEBHOOK_SECRET`).
- IP Whitelisting:** Validate incoming headers to ensure origin matches IntaSend production servers.
- Idempotency Check:** Check incoming `invoice_id` or `tracking_id` against recorded transactions in database to prevent double-processing.

4.2 EXPECTED WEBHOOK PAYLOAD SCHEMA

```
{ "invoice_id": "INV-INUABIZ-88210", "state": "COMPLETE", "provider": "MPESA", "charges": "0.00",
"net_amount": "3000.00", "currency": "KES", "account": "254712345678", "api_ref":
"tenant_uuid_94812301", "host": "https://api.inuabiz.co.ke", "challenge": "7f8b92c10a3e4d56" }
```

5. Database Schema for IntaSend Transactions

Column Name	Data Type	Constraints	Description
id	UUID		Unique record identifier.

Column Name	Data Type	Constraints	Description
		PRIMARY KEY, DEFAULT gen_random_uuid()	
tenant_id	UUID	FOREIGN KEY -> tenants(id)	References the vendor account.
invoice_id	VARCHAR(100)	UNIQUE, NOT NULL	IntaSend generated invoice ID.
amount	NUMERIC(10,2)	NOT NULL	Transaction amount (e.g., 3000.00).
payment_channel	VARCHAR(50)	NOT NULL	'MPESA_STK', 'CARD', or 'PAYBILL'.
status	VARCHAR(30)	NOT NULL	'PENDING', 'COMPLETE', 'FAILED'.
raw_webhook_payload	JSONB	NULLABLE	Full raw payload stored for auditing.
created_at	TIMESTAMPTZ	DEFAULT NOW()	Timestamp of transaction initiation.

6. Unclaimed & Failed Payment Handling

In high-volume mobile networks, edge cases like network timeouts, user cancellation, or bad phone inputs can occur. InuaBiz incorporates a multi-layer fallback strategy:

- **Failed Pin / Timeout:** System displays immediate UI alert with a retry button and alternative Paybill / Till instructions.
- **Unmapped Payment Webhook Queue:** If an STK payment completes but the `api_ref` (tenant ID) is missing or corrupted, the payload lands in the **Super-Admin Unclaimed Queue**. The platform admin can manually assign the transaction to the correct vendor in one click.
- **Automated Re-Check Cron:** A scheduled Supabase background worker polls IntaSend API for any transactions stuck in PENDING status for longer than 3 minutes to resolve edge-case delays.